

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

DECEMBER/JANUARY EXAMINATION

AMMS2613 PROBABILITY AND STATISTICS

TUESDAY, 6 JANUARY 2026

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMMS2613 PROBABILITY AND STATISTICS**Question 1**

- a) The following table shows the running time, measured to the nearest hour, of the battery for a random sample of 46 different laptops.

Battery Life (hours)	Number of Laptops
6 – 8	5
9 – 11	6
12 – 14	10
15 – 17	14
18 – 20	7
21 – 23	4

- (i) Calculate the mean, the variance, and the standard deviation. (9 marks)
- (ii) Calculate the mode. (4 marks)
- (iii) Construct a less than cumulative frequency distribution. (3 marks)
- (iv) Construct a less than cumulative frequency polygon and use it to estimate the median. (6 marks)
- b) Classify each of the following variables as a qualitative variable, a discrete quantitative variable, or a continuous quantitative variable.
- (i) Distance a runner covers in a marathon. (1 mark)
- (ii) Methods of learning. (1 mark)
- (iii) Number of invoices processed by an accountant. (1 mark)

[Total: 25 marks]

Question 2

- a) A market survey indicates that the probability a customer will rate a new product as excellent is 0.36. If a random sample of 9 customers who bought the product is chosen, find the probability that
- (i) less than 2 of them rate the product as excellent; (3 marks)
- (ii) at least 3 of them rate the product as excellent. (3 marks)

AMMS2613 PROBABILITY AND STATISTICS**Question 2 (Continued)**

- b) The useful life of a certain type of lithium battery is normally distributed with a mean of 1800 hours and a standard deviation of 120 hours. If a battery is selected randomly, find the probability that it has a useful life of
- (i) more than 1950 hours; (3 marks)
 - (ii) less than 1590 hours; (4 marks)
 - (iii) between 1700 hours and 1920 hours. (4 marks)
- c) A large computer system experiences random crashes that follow a Poisson distribution with a mean of 2 crashes per year. Find the probability that over a randomly selected year, the system crashes
- (i) two or three times; (4 marks)
 - (ii) more than two times. (4 marks)

[Total: 25 marks]

Question 3

- a) A survey of a random sample of 80 university students revealed that the average time they spend commuting to campus is 48 minutes. The population standard deviation of the commute times is 16 minutes. Construct a 95% confidence interval for the true population mean commute time for all university students. (6 marks)
- b) An airline states that the average weight of checked luggage is 20.5 kg. A baggage handler believes that the true average is more than 20.5 kg, potentially due to new travel trends. A random sample of 60 checked luggage yields a mean weight of 22 kg with a standard deviation of 5.8 kg. Can the baggage handler's belief be supported? Test at the 5% level of significance. (9 marks)
- c) A marketing campaign aimed to raise product awareness, expecting that 65% of consumers would recognise the brand name afterward. A post-campaign survey of 150 randomly selected consumers found that 84 recognised the brand name. Test the hypothesis at the 10% significance level that the true awareness rate is less than 65%. (10 marks)

[Total: 25 marks]

AMMS2613 PROBABILITY AND STATISTICS**Question 4**

- a) A local council surveyed 400 residents regarding their home type (Apartment or House) and whether they own a pet (Yes or No).

Pet Ownership	Apartment (A)	House (H)	Total
Yes (Y)	56	115	171
No (N)	154	75	229
Total	210	190	400

If a resident is selected randomly, find the probability that the resident

- (i) lives in a house and does not own a pet; (2 marks)
 - (ii) does not own a pet; (2 marks)
 - (iii) lives in a house or owns a pet; (3 marks)
 - (iv) lives in an apartment given that he/she does not own a pet. (3 marks)
-
- b) An operations manager analyses the relationship between the age of a machine (X) (in years) and the annual maintenance cost (Y) (in RM hundred) for 7 identical machines.

Machine	Age (Years), X	Cost (RM00), Y
A	3	8
B	6	18
C	4	16
D	8	21
E	12	26
F	10	24
G	15	27

- (i) Calculate the product moment correlation coefficient and interpret the result. (7 marks)
- (ii) Find the least squares regression equation of Y on X . (5 marks)
- (iii) Estimate the annual maintenance cost for a machine that is 9 years old. Is the prediction accurate and reliable? Explain. (3 marks)

[Total: 25 marks]

AMMS2613 PROBABILITY AND STATISTICS**FORMULAE****Mode, grouped data**

$$\text{Mode} = L_m + \frac{(f_m - f_b)}{2f_m - (f_b + f_a)} \times c_m$$

Sample mean, raw data

$$\bar{X} = \frac{\sum x}{n}$$

Sample mean, grouped data

$$\bar{X} = \frac{\sum fx}{\sum f}$$

Population standard deviation, raw data

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}} \quad \text{or} \quad \sqrt{\frac{\sum x^2}{N} - \mu^2}$$

Sample standard deviation, raw data

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} \quad \text{or} \quad \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$$

Population standard deviation, grouped data

$$\sigma = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

Sample standard deviation, grouped data

$$s = \sqrt{\frac{\sum fx^2 - \frac{(\sum fx)^2}{\sum f}}{\sum f - 1}}$$

Test Statistic for One Population

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

$$Z = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \quad (\text{for big sample})$$

$$Z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Median, grouped data

$$\text{Median} = L_m + \frac{c_m}{f_m} \left(\frac{n}{2} - \sum f_{m-1} \right)$$

Product moment coefficient of correlation

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Spearman's rank correlation coefficient

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Least squares regression line:

$$y = a + bx$$

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$a = \frac{\sum y}{n} - b \frac{\sum x}{n}$$

Binomial Probability Function

$$P(X = x) = {}^n C_x p^x (1-p)^{n-x}; \quad x = 0, 1, \dots, n$$

Poisson Probability Function

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad \lambda > 0, \quad x = 0, 1, 2, \dots$$

Confidence Interval for One Population

$$\bar{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

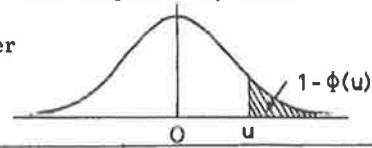
$$\bar{X} \pm Z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \quad (\text{for big sample})$$

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

AMMS2613 PROBABILITY AND STATISTICS**AREAS IN TAIL OF THE NORMAL DISTRIBUTION**

The function tabulated is $1 - \Phi(u)$ where $\Phi(u)$ is the cumulative distribution function of a standardised Normal variable u . Thus $1 - \Phi(u) = \frac{1}{\sqrt{2\pi}} \int_u^{\infty} e^{-u^2/2} du$ is the probability that a

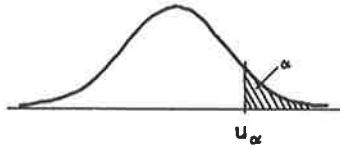
standardised Normal variable selected at random will be greater than a value of $u \left(= \frac{x - \mu}{\sigma} \right)$



$\frac{(x - \mu)}{\sigma}$	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
3.0	.00135									
3.1	.00097									
3.2	.00069									
3.3	.00048									
3.4	.00034									
3.5	.00023									
3.6	.00016									
3.7	.00011									
3.8	.00007									
3.9	.00005									
4.0	.00003									

AMMS2613 PROBABILITY AND STATISTICS**PERCENTAGE POINTS OF THE NORMAL DISTRIBUTION**

The table gives the 100α percentage points, u_α , of a standardised Normal distribution where $\alpha = \frac{1}{\sqrt{2\pi}} \int_{u_\alpha}^{\infty} e^{-u^2/2} du$. Thus u_α is the value of a standardised Normal variate which has probability α of being exceeded.



α	u_α	α	u_α	α	u_α	α	u_α	α	u_α	α	u_α
.50	0.0000	.050	1.6449	.030	1.8808	.020	2.0537	.010	2.3263	.050	1.6449
.45	0.1257	.048	1.6646	.029	1.8957	.019	2.0749	.009	2.3656	.010	2.3263
.40	0.2533	.046	1.6849	.028	1.9110	.018	2.0969	.008	2.4089	.001	3.0902
.35	0.3853	.044	1.7060	.027	1.9268	.017	2.1201	.007	2.4573	.0001	3.7190
.30	0.5244	.042	1.7279	.026	1.9431	.016	2.1444	.006	2.5121	.00001	4.2649
.25	0.6745	.040	1.7507	.025	1.9600	.015	2.1701	.005	2.5758	.025	1.9600
.20	0.8416	.038	1.7744	.024	1.9774	.014	2.1973	.004	2.6521	.005	2.5758
.15	1.0364	.036	1.7991	.023	1.9954	.013	2.2262	.003	2.7478	.0005	3.2905
.10	1.2816	.034	1.8250	.022	2.0141	.012	2.2571	.002	2.8782	.00005	3.8906
.05	1.6449	.032	1.8522	.021	2.0335	.011	2.2904	.001	3.0902	.000005	4.4172